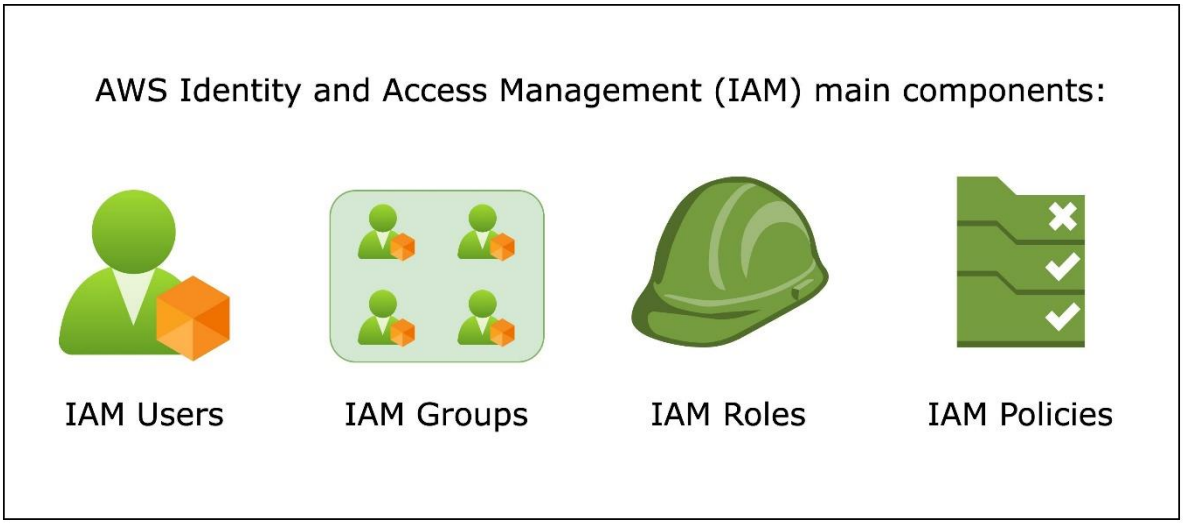
AWS

Date:29-10-2026

1.IAM Users, Roles, Policies, and Trust Relationships:

What is AWS IAM:

AWS IAM (Identity and Access Management) is a service that helps you control who can access your AWS resources and what actions they can perform.



1. IAM User

Represents an identity that needs access to AWS, often an individual person or an application requiring long-term credentials.

Example: A developer who needs to access AWS services.

2. IAM Role

Provides temporary permissions that a trusted user, AWS service, or other identity can assume.

Example: An EC2 instance assumes a role to read files from S3 without storing access keys on the instance.

3. IAM Policy

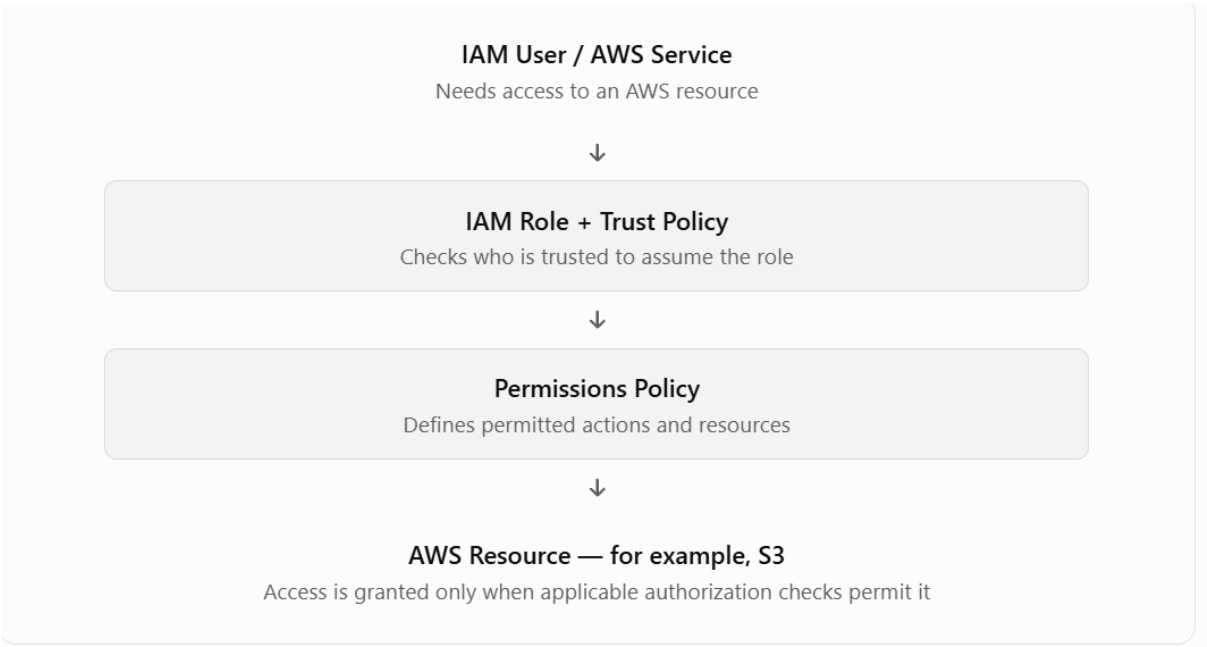
A JSON document that defines which actions are allowed or denied on which resources, optionally under specific conditions.

Example: Allow a user to list objects in a particular S3 bucket.

4. Trust Relationship

Defines who or what is allowed to assume an IAM role. It is specified in the role's trust policy.

Example: Allow the EC2 service to assume a particular IAM role.



2.Least-Privilege Access and Permission Management in AWS:

What is least privilege:

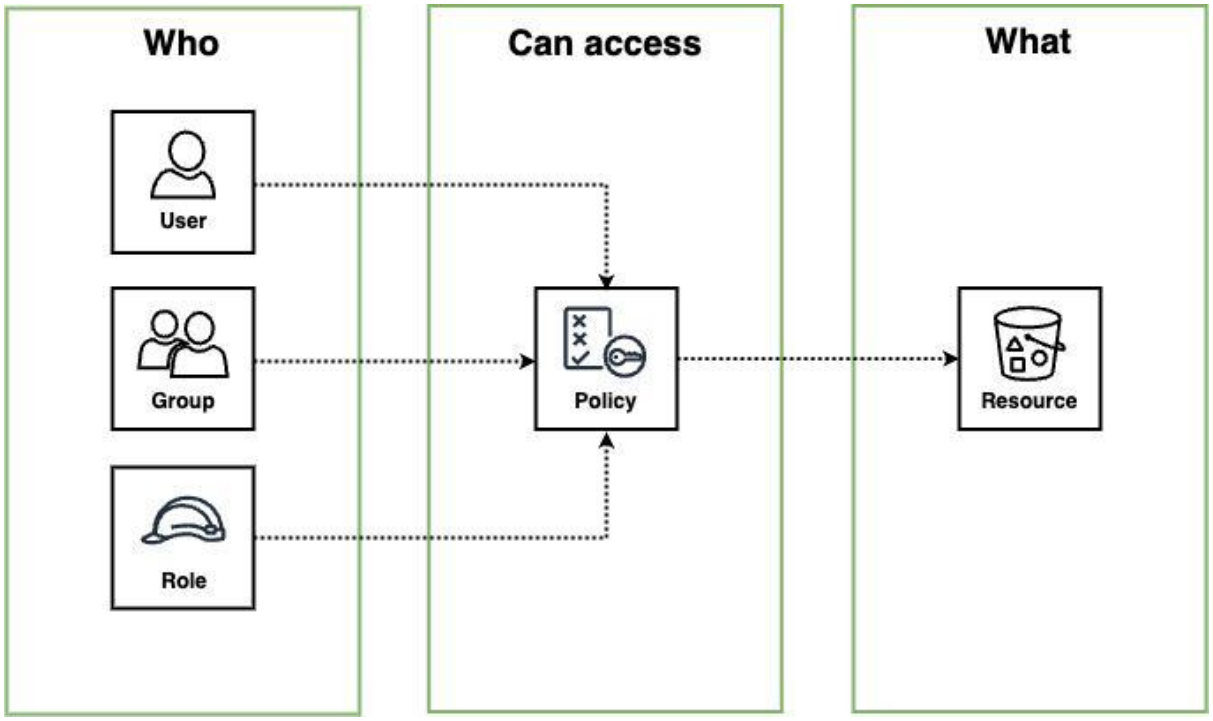
The principle of least privilege means giving a user, application, or AWS service the minimum permissions required to perform its task.

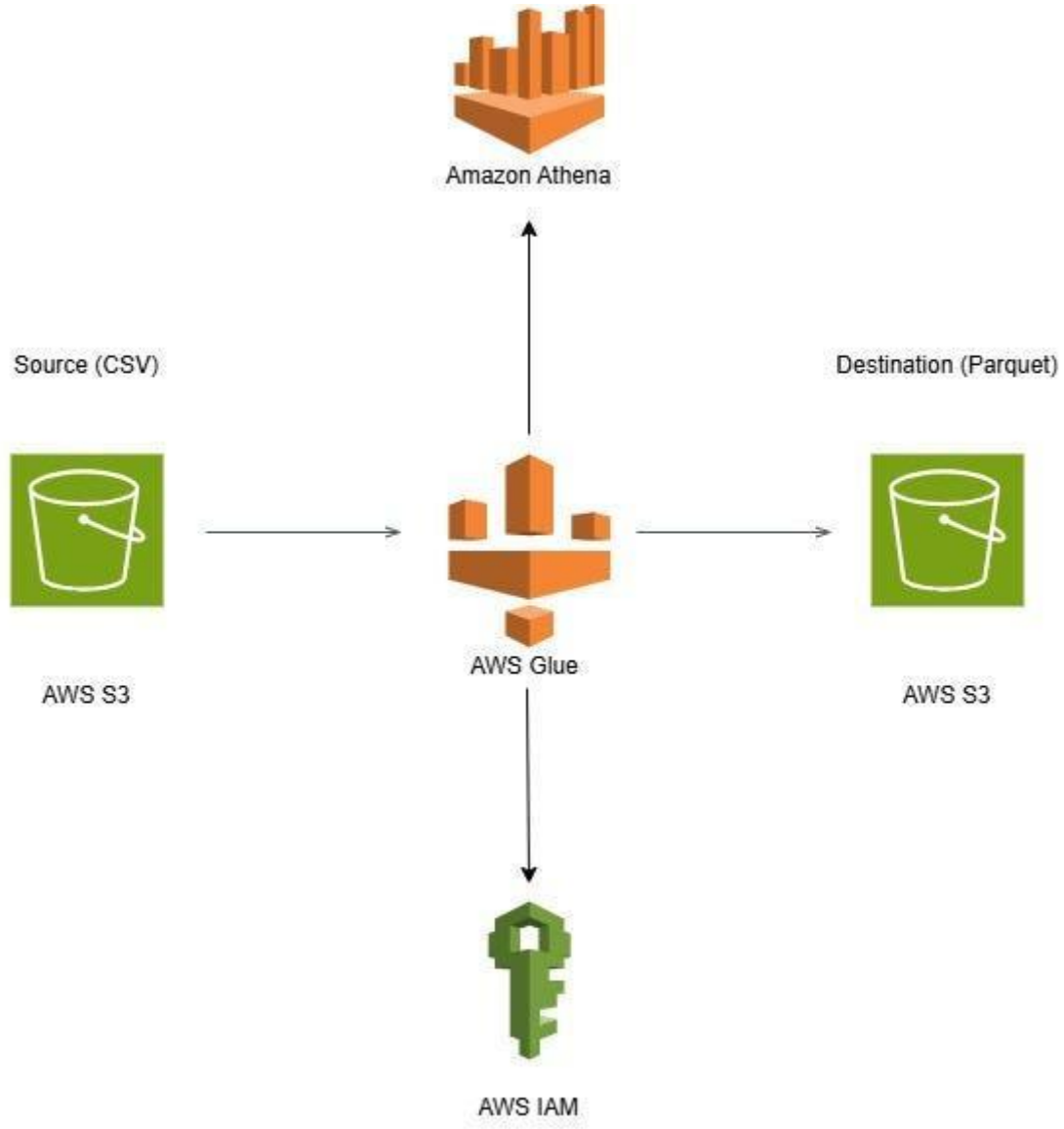
Understand permission management

AWS IAM permissions are primarily managed using policies.

Permission	AWS action example	Purpose
List bucket	`s3:ListBucket`	View objects within a bucket
Read object	`s3:GetObject`	Download or read an object
Upload object	`s3:PutObject`	Upload a file

Delete object	`s3:DeleteObject`	Delete a file





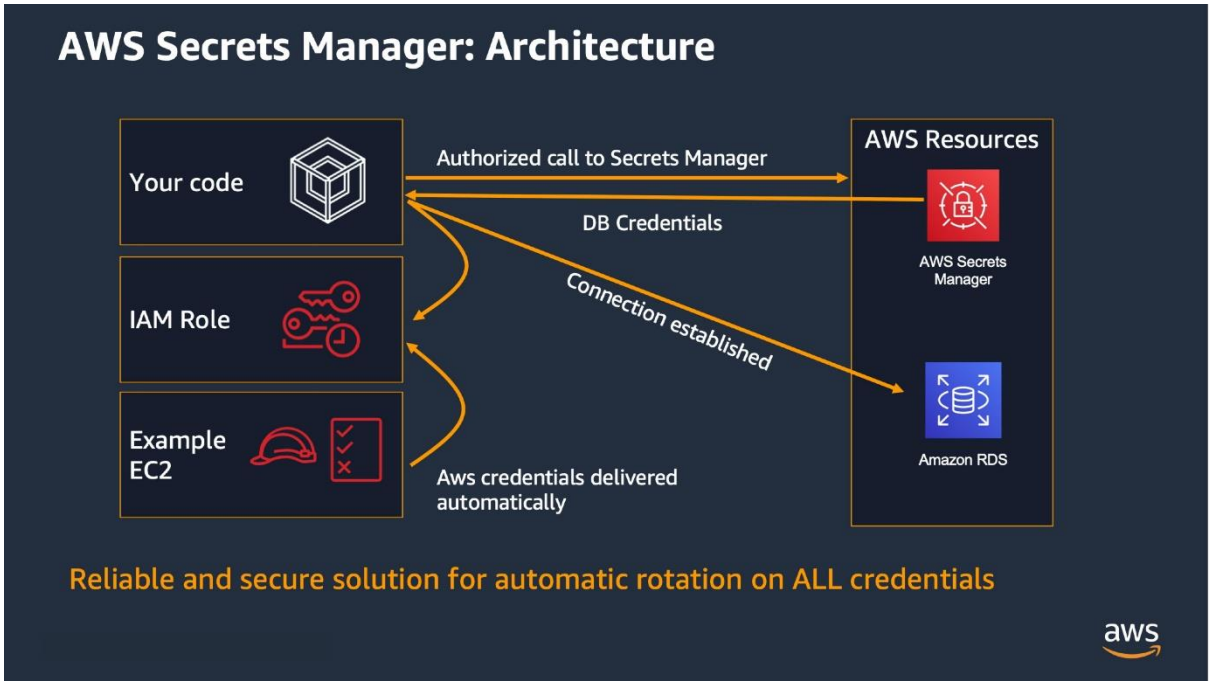
3.AWS Secrets Manager and AWS KMS:

1. What is AWS Secrets Manager?

AWS Secrets Manager is a service used to securely store, retrieve, and manage sensitive information called secrets.

Examples of secrets:

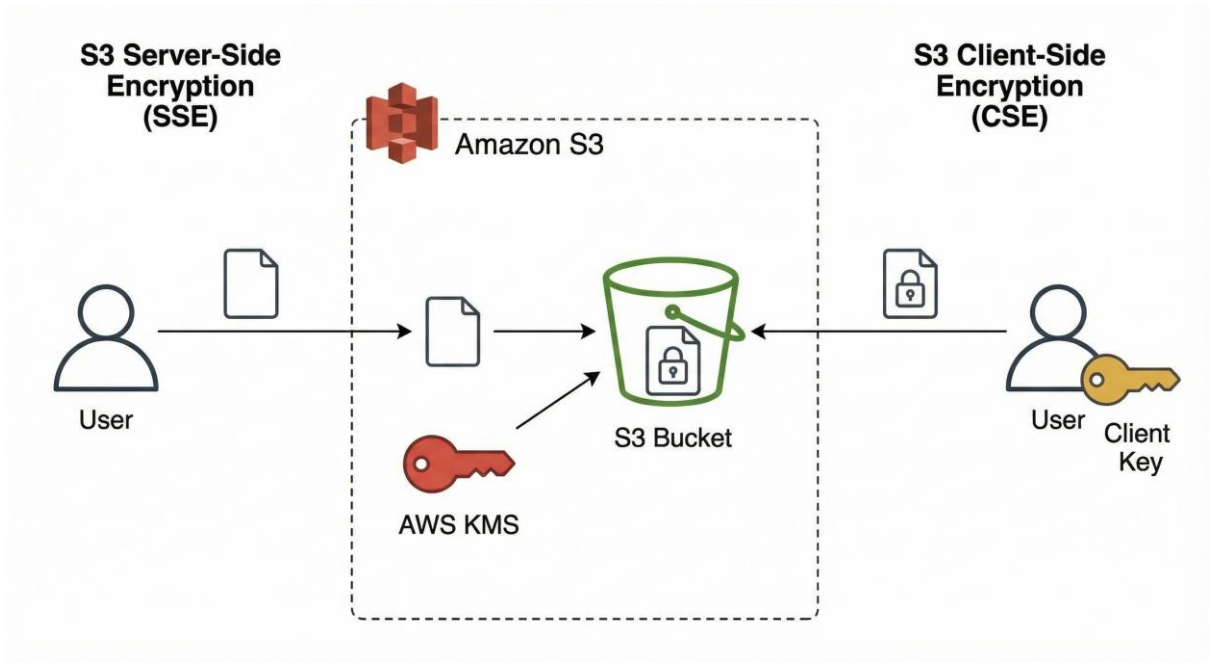
- Database usernames and passwords.
- API keys.
- Application credentials.
- Authentication tokens.



2. What is AWS KMS?

AWS KMS stands for AWS Key Management Service.

It is used to create and manage cryptographic keys that protect data through encryption and decryption.



Important KMS concepts:

Concept	Meaning
KMS Key	Cryptographic key used for encryption operations
Encryption	Converts readable data into unreadable ciphertext
Decryption	Converts ciphertext back into readable data
Customer managed key	Key that you create and manage in KMS
AWS managed key	Key managed by AWS for a particular service

3. Secrets Manager vs KMS:

Feature	Secrets Manager	AWS KMS
Main purpose	Store sensitive information	Manage encryption keys
Stores passwords	Yes	Not intended for this
Stores API keys	Yes	Not intended for this
Encryption	Uses encryption, commonly through KMS	Provides cryptographic operations
Automatic rotation	Supports secret rotation	Supports automatic key rotation for eligible keys

3.Amazon S3 Buckets, Prefixes, and Object Storage:

1. What is Amazon S3?

Amazon S3 (Simple Storage Service) is an AWS object storage service used to store and retrieve files and data over the internet.

You can store:

- CSV files
- JSON files
- Excel files
- Images and videos
- Log files
- Parquet files
- Backup files



What is an S3 Bucket?

An S3 bucket is a container used to store objects (files).

What are S3 Prefixes?

A prefix is the beginning portion of an object's key that helps organize objects into logical groups.

S3 vs Traditional File Storage

Feature	Amazon S3	Traditional file storage
Storage type	Object storage	File storage
Organization	Bucket and object keys	Folders and files
Scalability	Highly scalable	Depends on storage system
Common usage	Data lakes, backups	Shared file systems
Access	API and HTTP	File system protocols

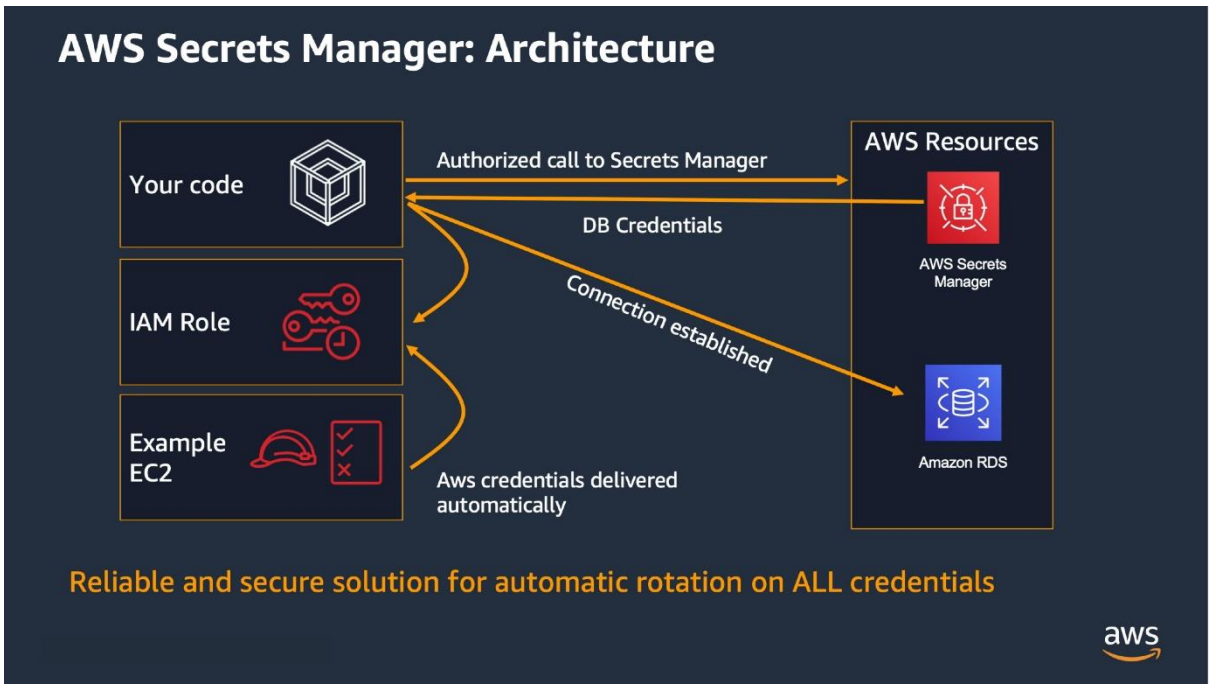
4.Secret Management and Key Rotation in AWS:

1. What is Secret Management?

Secret Management is the process of securely storing, accessing, controlling, and managing sensitive information.

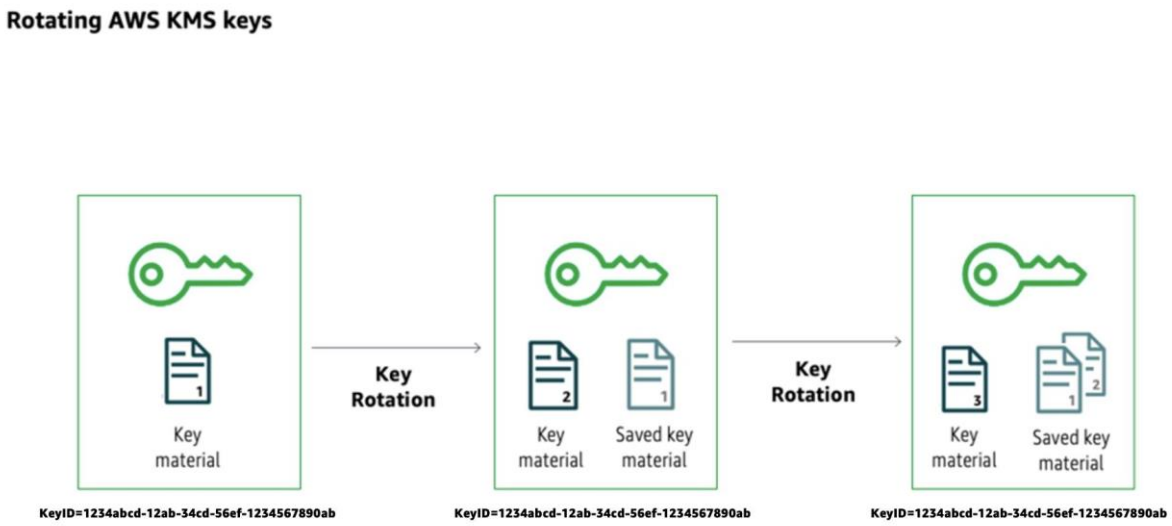
Examples of secrets:

- Database usernames and passwords.
- API keys.
- Application credentials.
- Access tokens.
- Authentication tokens.

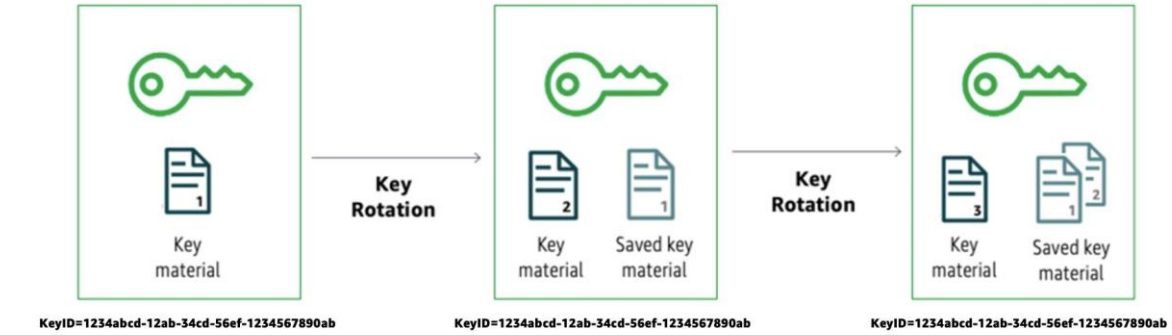


2. What is Key Rotation?

Key Rotation means periodically changing or replacing cryptographic keys to improve security.

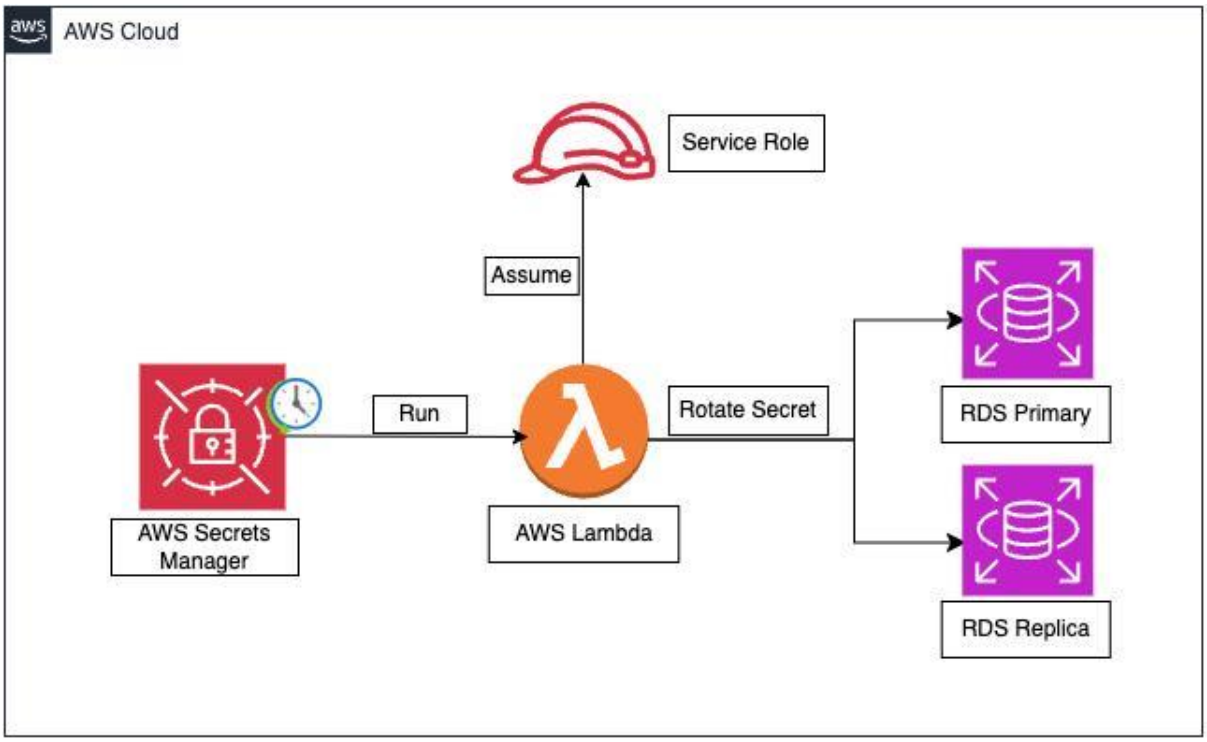


Rotating AWS KMS keys



3. What is Secret Rotation?

Secret rotation means periodically changing sensitive credentials, such as database passwords or API keys.



4. Secret Rotation vs Key Rotation

Feature	Secret Rotation	Key Rotation
Main purpose	Change credentials	Change cryptographic key material
AWS service	Secrets Manager	AWS KMS
Example	Database password	Encryption key
Frequency	Based on security requirements	Based on key policy and requirements
Automation	Rotation Lambda	KMS automatic rotation for supported keys

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Date:02-11-206

Amazon CloudWatch – Logs, Metrics, and Alarms

What is Amazon CloudWatch?

Amazon CloudWatch is an AWS monitoring service used to collect, monitor, analyze, and visualize operational data from AWS resources and applications.

CloudWatch helps you monitor:

- CPU utilization.

- Memory usage (with an appropriate agent or custom metrics).
- Application errors.
- Server performance.
- Network activity.
- Application logs.

Three important components of CloudWatch:

What are CloudWatch Logs?

CloudWatch Logs is a feature that collects, stores, and monitors log data from applications and AWS services.

What are logs?

Logs are records of events that happen inside an application or system.

Important CloudWatch Logs concepts:

Concept	Meaning
Log event	An individual record of an event
Log stream	Sequence of log events from one source
Log group	Collection of related log streams
Retention	How long logs are stored
Log Insights	Tool to query and analyze logs

What are CloudWatch Metrics?

CloudWatch Metrics are numerical measurements used to monitor the performance and health of AWS resources.

Metric	Meaning
CPU Utilization	Percentage of CPU being used
Memory Usage	Amount of memory being used
Network In	Incoming network traffic
Network Out	Outgoing network traffic
Disk Usage	Storage utilization
Request Count	Number of application requests

What are CloudWatch Alarms?

CloudWatch Alarms monitor metrics and perform actions when specified conditions are met.

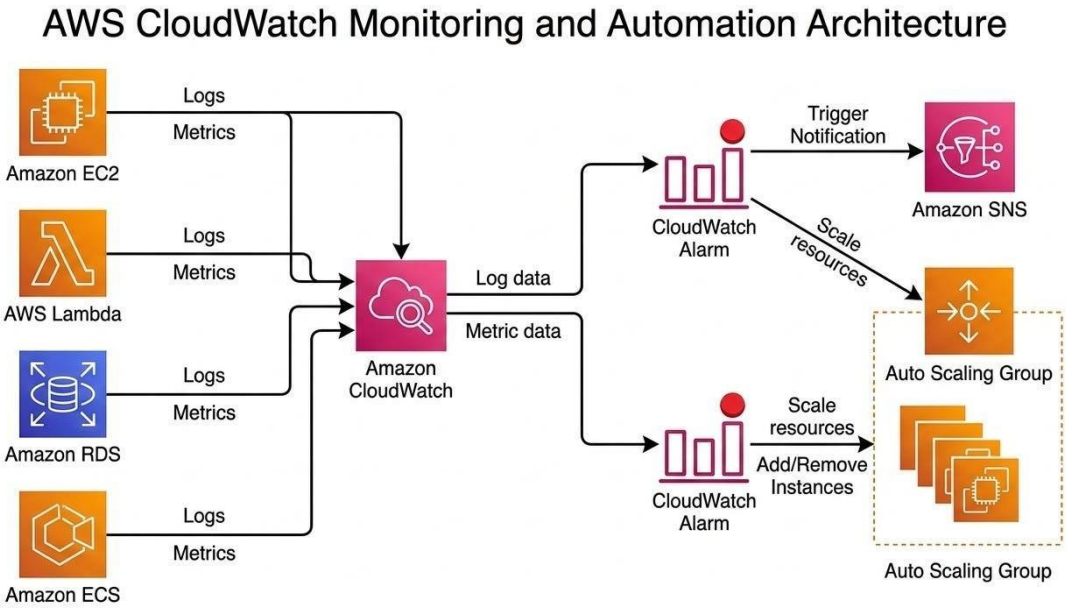
Three important alarm states:

State	Meaning
OK	Metric is within the configured threshold
ALARM	Metric has crossed the configured threshold
INSUFFICIENT_DATA	Not enough data to determine the state

Types of CloudWatch Alarms:

1. Metric Alarm – Monitors one or more metrics.
2. Composite Alarm – Combines multiple alarm conditions.
3. Anomaly Detection Alarm – Uses expected metric patterns to identify unusual behavior.

Amazon CloudWatch working Architecture:



Important AWS services connected with CloudWatch:

AWS Service	Purpose
Amazon EC2	Monitor virtual servers
Amazon S3	Monitor storage metrics and request activity
Amazon RDS	Monitor database performance
AWS Lambda	Monitor function execution and errors
Amazon SNS	Send notifications
AWS Auto Scaling	Adjust resource capacity

Difference between Logs, Metrics, and Alarms

Feature	Logs	Metrics	Alarms
Purpose	Record events	Measure performance	Detect conditions
Data type	Text/events	Numerical values	Monitoring state
Example	Database error	CPU = 85%	CPU above 80%
Main use	Troubleshooting	Performance analysis	Notifications/actions

AWS Step Functions and State Machines:

What is AWS Step Functions?

AWS Step Functions is a fully managed AWS service used to coordinate multiple AWS services into workflows.

For example, imagine you are building a data pipeline:

1. Read data from Amazon S3.
2. Validate the data.
3. Process the data using AWS Glue.
4. Store the processed data in Amazon S3.
5. Send a notification when processing is completed.

Instead of manually managing every step, AWS Step Functions coordinates the entire workflow.

2. Why do we need AWS Step Functions?

Imagine you have a data engineering project with multiple tasks.

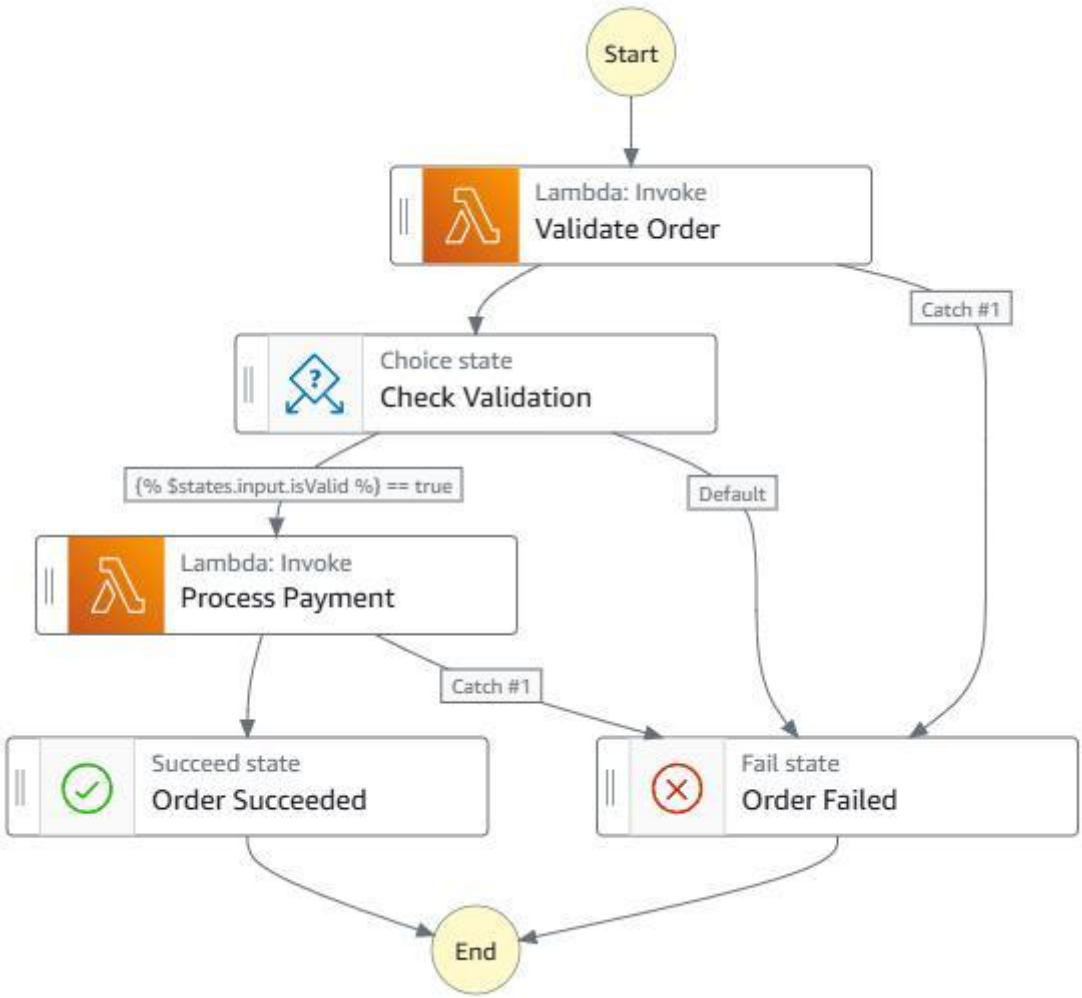
Without Step Functions:

- You need to manage task execution.
- You need to handle failures.
- You need to manage task dependencies.
- You need to coordinate different AWS services.
- You need to manage retries manually in your application.

With Step Functions:

- Automates workflow execution.
- Connects multiple AWS services.
- Handles task failures.
- Supports retries and error handling.
- Tracks workflow execution.
- Supports parallel processing.

AWS Step Functions works:



What is a State Machine?

A State Machine is the definition of a workflow that describes:

- What tasks should execute.
- In which order tasks should execute.
- What happens when a task succeeds.
- What happens when a task fails.
- Which conditions control the workflow.

In AWS Step Functions, a state machine defines the complete workflow using Amazon States Language (ASL), a JSON-based language.

types of States in AWS Step Functions

AWS Step Functions provides different state types to control workflow execution.

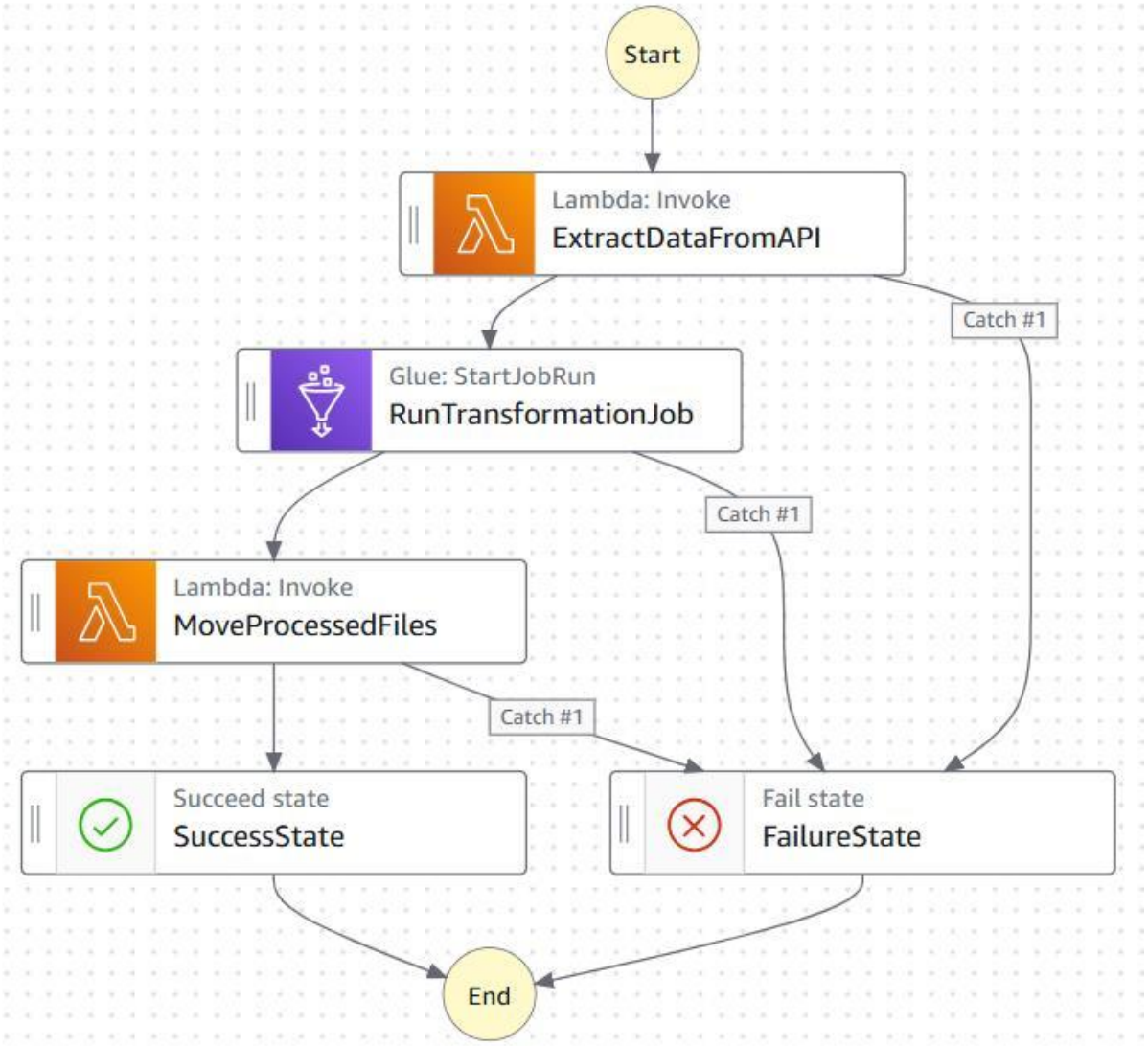
State	Purpose
Task	Performs a unit of work
Choice	Makes decisions based on conditions
Pass	Passes input to output without performing work
Wait	Pauses workflow execution
Parallel	Executes multiple branches simultaneously
Map	Processes multiple items or records
Succeed	Successfully ends a workflow
Fail	Ends a workflow with failure

A. Task State

A Task State performs an actual operation.

Example:

- Execute an AWS Glue job.
- Invoke a Lambda function.
- Call another supported AWS service integration.



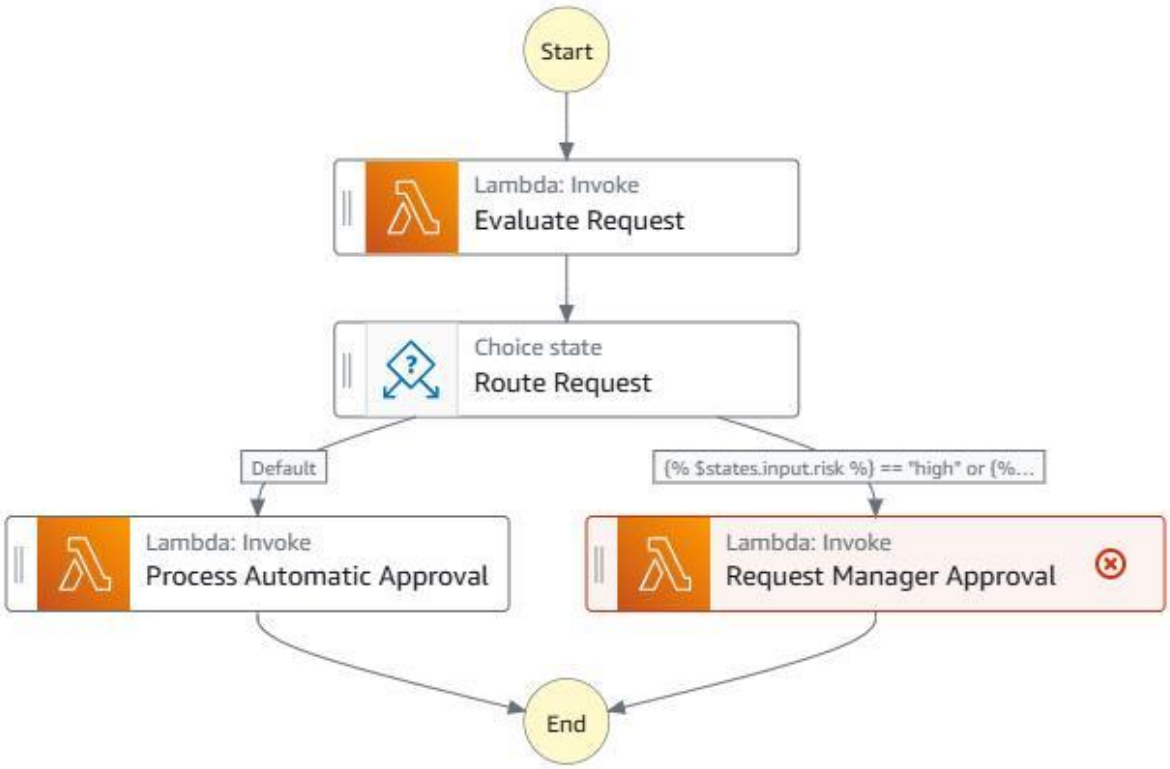
B. Choice State

A Choice State makes decisions based on conditions.

Example:

Suppose you are validating customer records.

- If records are valid → Continue processing.
- If records are invalid → Send them for error handling.



C. Pass State

A Pass State passes input to the next state without performing a task.

D. Wait State

A Wait State pauses workflow execution for a specified period or until a specified time.

Example:

Wait 5 minutes before checking the next processing stage.

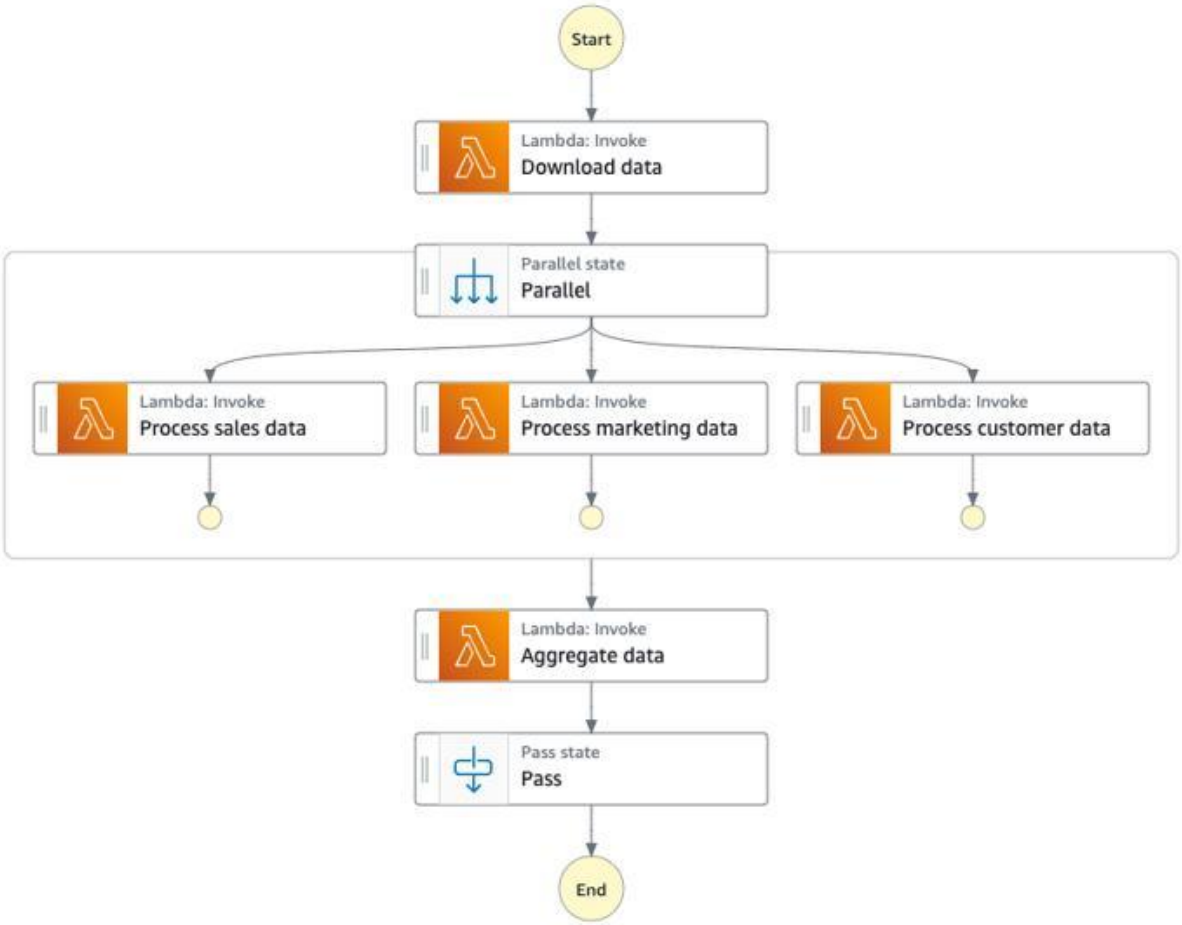
E. Parallel State

A Parallel State executes multiple branches of a workflow at the same time.

Example:

After receiving customer data:

- Branch 1: Validate customer details.
- Branch 2: Check customer transactions.
- Branch 3: Analyze customer activity.



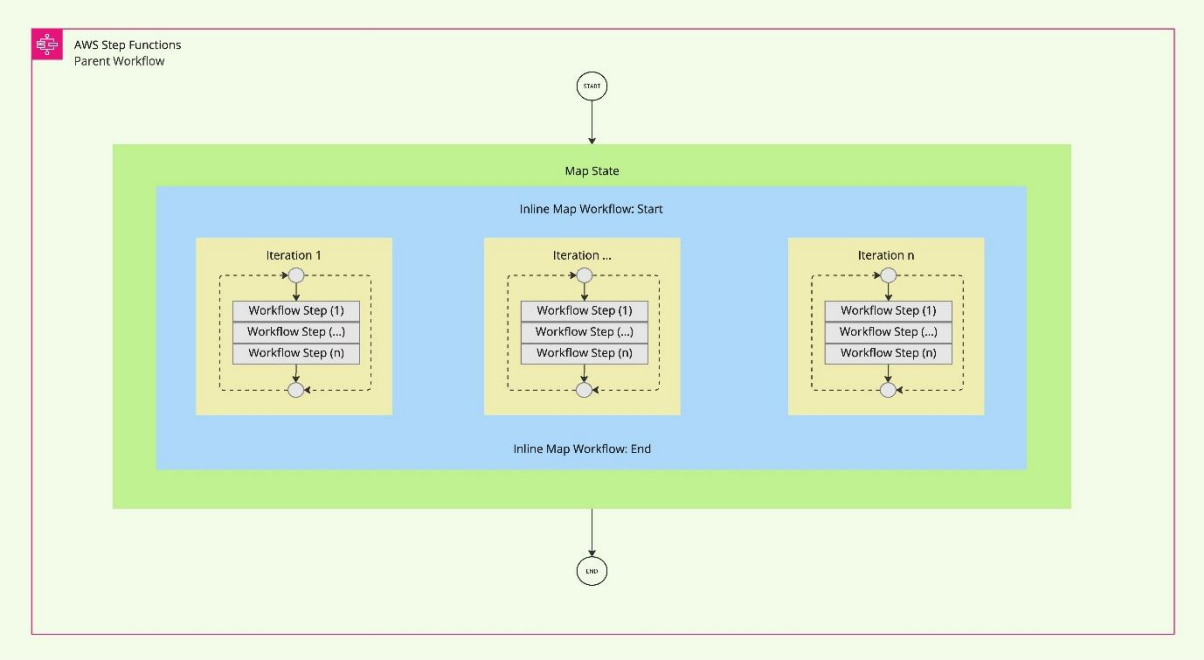
F. Map State

A Map State processes multiple items using the same set of steps.

Example:

Suppose you have 1,000 customer records.

Instead of processing every record with separate workflow definitions, a Map State can apply the same processing steps to each item.



G. Succeed and Fail States

- Succeed State: Marks the workflow as successfully completed.
- Fail State: Stops the workflow and reports failure.

These states help control the final outcome of a workflow.